

## **Sophie ARHEIX-PARRAS, Ph.D., SLP**

Postdoctoral Researcher at University of South Carolina

Incoming Assistant Professor at UT Dallas (2027)

Department of Psychology, University of South Carolina,  
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### **EDUCATION**

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| Oct 2020 - Oct 2023   | <b>Ph.D. in Cognitive Science</b><br><i>University of Bordeaux (France)</i><br>Transcranial Magnetic Stimulation for Post-Stroke Aphasia:<br>Individualization and Electrophysiological Changes <ul style="list-style-type: none"><li>• ACTIVE Lab, U1219, INSERM, Bordeaux Population Health</li><li>• Supervisor: Prof. Glize (Bordeaux, France), Co-supervisor: Dr. Python (Geneva, Switzerland),</li><li>• Examiners: Prof. Guggisberg (Bern, Switzerland), Prof. de Boissezon (Toulouse, France), Prof. Hesling (Bordeaux, France)</li></ul> 5-month placement in the NeuroPsychoLinguistic Lab, University of Geneva, Switzerland (Director Prof. Laganaro) |
| Sept 2019 - Sept 2020 | <b>Master's in Neuroscience and Clinical Neuropsychology</b><br><i>Universities of Toulouse, Grenoble &amp; Lyon (France)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Sept 2016 - June 2020 | <b>Bachelor's &amp; Master's in Speech-Language Pathology</b><br>2016-2020: <i>University of Bordeaux (France)</i><br>2015-2016: <i>University of Nancy (France)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

### **ACADEMIC POSITION**

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| Incoming – Fall 2027 | <b>Assistant Professor</b><br><i>University of Texas at Dallas, TX (USA)</i><br>Department of Speech, Language and Hearing<br>School of Behavioral and Brain Sciences |
| Feb 2024 - present   | <b>Postdoctoral Researcher</b><br><i>University of South Carolina, Columbia, SC (USA)</i>                                                                             |

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|                      | Lexicosemantic processes of language in neurotypical individuals and people with post-stroke aphasia<br>Supervisor: Prof. Desai |
| Sept 2022 - Aug 2024 | <b>Temporary teaching and research associate</b><br><i>University of Bordeaux (France)</i><br>Department of Health Science      |
| Sept 2021 - Aug 2022 | <b>Temporary teaching and research functions</b><br><i>University of Bordeaux (France)</i><br>Department of Health Science      |

## CLINICAL POSITION

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| Jan 2021 - Aug 2022  | <b>Speech and Language Pathologist</b><br><i>Bordeaux University Hospital Center (France)</i><br>Cognition & Language Unit<br>Assessments and telerehabilitation for people with acquired brain lesions, neurology unit                   |
| Nov 2020 - Dec 2020  | <b>Speech and Language Pathologist</b><br><i>Bordeaux University Hospital Center (France)</i><br>CAMPS<br>Assessments and rehabilitation for children with neurological disorders                                                         |
| July 2020 - Aug 2020 | <b>Speech and Language Pathologist</b><br><i>Bordeaux University Hospital Center (France)</i><br>Service of Physical Medicine and Rehabilitation<br>Assessments and rehabilitation for people with acquired brain lesions, neurology unit |

## PUBLICATIONS

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### International peer-reviewed scientific journals

- Siklafidou, I. P., **Arheix-Parras, S.**, Hameau, S., Python, G., Glize, B. (2026). Efficacy of rTMS in Post-Stroke Bilingual Aphasia: A Single Case Experimental Design. *International Journal of Language & Communication Disorders*. <https://doi.org/10.1111/1460-6984.70311>
- Xiao, C., **Arheix-Parras, S.**, Riccardi, N., Johari, K., Desai, R. H. (2026). Anterior Temporal Lobe Mediates Associative and Taxonomic Neighborhood Effects in Lexical Processing. *Cortex*. [10.1016/j.cortex.2026.06.026](https://doi.org/10.1016/j.cortex.2026.06.026)

- Arheix-Parras, S.,** Xiao, C., Crouse, S., Riccardi, N., Johari, K., Desai, R. H. (2026). The Role of Anterior Temporal Lobe in Reading: a tDCS Study. *Neurobiology of language*. <https://doi.org/10.1162/NOL.a.266>
- Arheix-Parras, S.,** Moore, S. R. & Desai, R. H. (2026). Improving Lexicosemantic Impairments in People with Post-stroke Aphasia using Repetitive Transcranial Magnetic Stimulation Targeting the Right Anterior Temporal Lobe. *Brain Science*. <https://doi.org/10.3390/brainsci16010117>
- Brun, C., **Arheix-Parras, S.,** Carlsberg, M., Glize, B., Cogné, M. (2026). Efficacy of telerehabilitation for cognitive impairments after brain lesions: a systematic review. *Annals of Physical and Rehabilitation Medicine*. <https://doi.org/10.1016/j.rehab.2025.102058>
- Python, G., Henchoz, M., Séchaud, E., Simao, L., **Arheix-Parras, S.,** Glize, B. (2025). Health-related quality of life in French-speaking persons with chronic aphasia: adaptation and validation of the SAQOL-39F. *Aphasiology*. <https://doi.org/10.1080/02687038.2025.2508448>
- Arheix-Parras, S.,** Franco, J., Siklafidou, I-P., Villain, M., Rogue, C., Python, G. & Glize, B. (2024) Neuromodulation of the Motor Cortex of the Lips Using rTMS to Improve Post-Stroke Aphasia Symptoms: a Single Case Experimental Design, *American Journal of Speech-Language Pathology*. [https://doi.org/10.1044/2024\\_AJSLP-23-00215](https://doi.org/10.1044/2024_AJSLP-23-00215)
- Arheix-Parras, S.,** De Puy De Goyne, M., Franco, J., Villain, M., Glize, B. & Python G. (2024) Transcranial Magnetic Stimulation to Improve Aphasia Following Right Hemispheric Stroke: a Single Case Experimental Design Study. *Annals of Physical and Rehabilitation Medicine*. <https://doi.org/10.1016/j.rehab.2024.101858>
- Arheix-Parras, S.,** Glize, B., Guehl, D., Python, G. (2023) Electrophysiological Modifications in Post-Stroke Aphasia: a Systematic Review. *Brain Topography*. <https://doi.org/10.1007/s10548-023-00941-4>
- Arheix-Parras, S.,** Barrios, C., Python, G., Cogné, M., Sibon, I., Engelhardt, M., Dehail, P., Cassoudeulle, H., Moucheboeuf, G. & Glize, B. (2021) A Systematic Review of Repetitive Transcranial Magnetic Stimulation in Aphasia Rehabilitation: Leads for Future Studies. *Neuroscience and Biobehavioral Reviews*. <https://doi.org/10.1016/j.neubiorev.2021.04.008>
- Carlsberg, M., Percey, V., **Arheix-Parras, S.,** Charaire, P., Cogné, M., Dehail, P., De Seze, M., Prouteau, A., Sibon, I., & Glize, B. (2020) Disability of People with Aphasia is Overestimated by General Practitioners. *Annals of Physical and Rehabilitation Medicine*. <https://doi.org/10.1016/j.rehab.2020.02.003>

## Work in progress

- Yang, Y., Newman-Norlund, R., Guan, X., Ahmadi, S., Willis, R., Salman, N., Warren, K., **Arheix-Parras, S.**, Nelakuditi, S., Bonilha, L., Rorden, C., Desai, R.H., Fridriksson, J. (*under review*). Recovering Lesion Parameters from Aphasic Picture Naming Error Profiles in Large Language Models. *arXiv preprint arXiv:2608.06429*.
- Yang, Y., Guan, X., **Arheix-Parras, S.**, Ahmadi, S., Newman-Norlund, R., Bonilha, L., Rorden, C., Fridriksson, J., Desai, R.H. & Nelakuditi, S. (*under review*). Lesioned Multimodal Language Models Reproduce Aphasic Picture-Naming Patterns. *arXiv preprint arXiv:2607.11621*.
- Rabastain, L., Vildy, M., **Arheix-Parras, S.** (*to be submitted*). Telerehabilitation of Post-Stroke Aphasia in Africa with Community Health Agents: a Feasibility and Acceptability study.
- Arheix-Parras, S.**, Sanchez, M., Franco, J., Villain, M., Python, G. & Glize, B. (*to be submitted*). Transcranial Magnetic Stimulation for People with Severe Post-Stroke Aphasia.
- Riccardi, N., **Arheix-Parras, S.**, Johari, K., Rorden, C., Fridriksson, J. & Desai, R. H. (*in preparation*). Lexical and Semantic Impairments After Anterior Temporal and Temporoparietal Lesions.
- Arheix-Parras, S.**, Xiao, C., Singh, S., Riccardi, N., Johari, K. & Desai, R. H. (*in preparation*). Differential effect of tDCS on Auditory and Visual Lexicosemantic Processes in the Anterior Temporal Lobe.
- Arheix-Parras, S.**, Moore, S. R., Rorden, C., Bonilha, L., & Desai, R. H. (*in preparation*). Recovery from Post-Stroke Aphasia With rTMS Targeting left or right Anterior Temporal Lobe: study protocol of the ATLAS Clinical Trial

## National articles and educational chapters

- Jouberton, A. & **Arheix-Parras, S.** (2024) Group approach in speech and language therapy: a systematic review (in French). *Rééducation orthophonique / N°300-301*.
- Arheix-Parras, S.**, Carlsberg, M. & Glize B. (2023) Updates on Aphasia: How to Integrate Rehabilitation into Daily Life and Develop Support for Caregivers (in French). *Société de Neuropsychologie de Langue Française*.
- Arheix-Parras, S.**, Hesling, I. & Glize B. (2022) Impact of Brain Lesions on Language Networks & Predictive Factors and Contributions of Clinical, Imaging, and Electrophysiological Aspects (in French). *De Boeck*.

Glize, B., **Arheix-Parras, S.**, Caron, A. & Engelhardt, M. (2020) Updates and Review on Aphasia (in French). *Actualités en MPR / N° 3-4*.

## **PRESENTATIONS**

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### **Invited talks**

- 2026      Language Network Reorganization Following Anterior Temporal Lobe rTMS in Post-Stroke Aphasia: preliminary results of the ATLAS Trial, S. Arheix-Parras, *Translational Imaging Symposium*, University of South Carolina, Columbia, USA
- 2026      Enhancing language recovery after a stroke through anterior temporal lobe stimulation, S. Arheix-Parras, *Neuroscience retreat*, University of South Carolina, Columbia, USA
- 2026      The Interest of Neuromodulation for the Recovery of Post-Stroke Aphasia, S. Arheix-Parras, *Entretiens de Médecine Physique et de Réadaptation (EMPR)*, Montpellier, France
- 2025      Aphasia Research in Action, A. Vogel-Eyny, M. Steinberg Lowe & S. Arheix-Parras, “Ask the Expert” (webinar)
- 2025      Optimizing repetitive Transcranial Magnetic Stimulation for Post-Stroke Aphasia: Toward Tailored Neurorehabilitation, S. Arheix-Parras, *Institute of Mind and Brain*, Columbia, SC, USA
- 2024      How can repetitive Transcranial Magnetic Stimulation be Individualized for Post-Stroke Aphasia Recovery? S. Arheix-Parras, *C-Star lecture series*, Columbia, SC, USA.
- 2023      Individualization of Transcranial Magnetic Stimulation for Post-Stroke Aphasia Recovery, S. Arheix-Parras, *CLOR* (on-line conference).
- 2022      Non-Invasive Brain Stimulation and Cognition, S. Arheix-Parras & F. Pizzo, *Ecole pratique Bancaud Talairach*, Bordeaux, France.
- 2022      Prolonged post-COVID-19 Cognitive Symptoms in Adults: What Management? M. Carlsberg & S. Arheix-Parras, *Training: Aphasia and Acquired Cognitive Disorders*, Bordeaux, France.
- 2022      Aphasia: Novelties and New Approaches for our Patients, Adapted to the Contexts of 2022, B. Glize & S. Arheix-Parras, *Sofmer* (webinar).

## Conference Presentations

- 2026 **Poster:** Anterior Temporal Lobe Stimulation for Post-Stroke Aphasia Recovery using rTMS (ATLAS): a Randomized Clinical Trial. Arheix-Parras S., Moore S., Rorden, C., Bonilha, L., Desai, R. H. *Academy of Aphasia*, Liège, Belgium
- 2026 **Poster:** Language Network Reorganization Following Anterior Temporal Lobe rTMS in Post-Stroke Aphasia: Neuroimaging Protocol of the ATLAS Trial. Arheix-Parras S., Moore S., Rorden, C., Bonilha, L., Desai, R. H. *Society for the Neurobiology of Language 17th Annual Meeting*, Genève, Switzerland
- 2026 **Poster:** Anterior Temporal Lobe stimulation using rTMS for post-stroke aphasia recovery (ATLAS): a study protocol for a single-blinded randomized clinical trial. Arheix-Parras S., Moore S., Rorden, C., Bonilha, L., Desai, R. H. *Science of Aphasia*, Bordeaux, France
- 2025 **Poster:** Improving Lexicosemantic Impairments in People with Post-stroke Aphasia using Repetitive Transcranial Magnetic Stimulation Targeting the Right Anterior Temporal Lobe. Arheix-Parras. S., Crouse, S., Moore S., Desai, R. H. *Academy of Aphasia*, San Diego, USA
- 2025 **Poster:** The role of anterior temporal lobe in reading: an HD-tDCS study. Arheix-Parras, S., Xiao, C., Crouse, S., Riccardi, N., Johari, K., Desai, R. H. *Society for the Neurobiology of Language 17th Annual Meeting*, Washington D.C., USA.
- 2025 **Poster:** Repetitive Transcranial Magnetic Stimulation Targeting the Right Anterior Temporal Lobe to Improve Lexicosemantic Impairments in People with Post-stroke Aphasia. Sophie Arheix-Parras. S., Crouse, S., Moore S., Desai, R. H. *Society for the Neurobiology of Language 17th Annual Meeting*, Washington D.C., USA.
- 2025 **Poster:** Replicating Aphasic Picture Naming in Large Language Models Desai, R.H., Yang, Y., Guan, X., Arheix-Parras, S., Nelakuditi, S. *Society for the Neurobiology of Language 17th Annual Meeting*, Washington D.C., USA.
- 2024 **Poster:** Semantic Neighborhood Effects with Anterior Temporal Lobe Stimulation. Xiao, C., Arheix-Parras, S., Johari, K., Riccardi, N., Desai, R.H. *Society for the Neurobiology of Language 16th Annual Meeting*, Brisbane, Australia.
- 2023 **Oral Communication:** Neuromodulation of the Motor Cortex of the Lips Using rTMS to Improve Post-Stroke Aphasia Symptoms, Arheix-Parras, S., De Puy De Goyne, M., Franco, J., Villain, M., Glize, B. & Python G. Nice, *Science of Aphasia*, France.
- 2023 **Poster:** Transcranial Magnetic Stimulation in Aphasia Targeting the Right Motor Cortex of the Lips : Behavioral Improvements and Electrophysiological Changes in Post-Stroke Aphasia. Arheix-Parras, S., De Puy De Goyne, M., Franco, J., Villain, M., Glize, B. & Python G. *Academy of Aphasia*, Reading, United Kingdom.

- 2022 **Oral Communication:** Telecare for Cognitive Rehabilitation of Brain Damaged Patients: the Experience of the COLA Unit of the Bordeaux University Hospital, M. Carlsberg, S. Arheix-Parras, H. Cassoudessale, E. Sorita, N. Duclos, B. Glize, *SOFMER Congress*, Rennes, France.
- 2022 **Oral Communication:** Fatigue, Depression and Cognitive Disorders: the Covid-long Triptych? H. Lemistre, M. Engelhardt, S. Arheix-Parras, M. Carlsberg, B. Glize, *SOFMER Congress*, Rennes, France.
- 2022 **Oral Communication:** Transcranial Magnetic Stimulation in a Left-Handed Patient with Aphasia Following a Right Stroke: a SCED Protocol of Clinical and Electrophysiological Changes, S. Arheix-Parras, M. Du Puy de Goyne, G. Python, B. Glize, *SOFMER Congress*, Rennes, France.
- 2022 **Poster:** Transcranial Magnetic Stimulation in a Left-Handed Patient with Aphasia Following a Right Stroke: a SCED Design of Clinical and Electrophysiological Changes, S. Arheix-Parras, M. Du Puy de Goyne, G. Python, B. Glize, *Science of Aphasia*, Bordeaux, France.
- 2021 **Workshop:** Language and Cognitive Telerehabilitation, M. Carlsberg, S. Arheix-Parras, B. Glize, *SOFMER Congress*, Lille, France.
- 2021 **Poster:** Effects of rTMS on Temporal Processes During Naming Tasks in Chronic Patients with Aphasia, S. Arheix-Parras, G. Python, N. Sentucq, M. du Puy de Goyne, E. Amauger, B. Glize, *SOFMER Congress*, Lille, France.

## GRANTS AND AWARDS

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- 2026 **NIDCD-NIH K99 DC023302**  
- *'Recovery from Post-Stroke Aphasia With rTMS Targeting The Anterior Temporal Lobe: Interhemispheric Activation Changes And Lesion-Connectome-Based Predictive Outcomes'*, PI: S. Arheix-Parras, Principal mentor: R.H.Desai, Co-mentors: C. Rorden & L. Bonilha (\$292,008)
- 2026 **NIDCD-NIH Travel Fellowship**  
Academy of Aphasia travel award (\$1,000)
- 2024 **Barbara Martin Aphasia grant**, co-sponsored by the **Academy of Aphasia**  
*'Anterior Temporal Lobe stimulation using Transcranial Magnetic Stimulation to modulate lexicosemantic processes'*, PI: S. Arheix-Parras (\$10,000)
- 2024 **AOI**, Bordeaux University Hospital (France)  
*'Individualization of rTMS in post-stroke aphasia based on the Disconnectome'*, Scientific manager: S. Arheix-Parras, PI: G. Chapouthier (35,492€)
- 2022 **UBGRS International Mobility**, University of Bordeaux (France): placement in Geneva, Switzerland.
- 2021 **Bordeaux Métropole** (France): support for the acquisition of a 128-channel EEG.

## TEACHING EXPERIENCE

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### **Aphasiology (2021-2024)**

- Assessment in aphasiology (Bachelor's degree, Speech Language Therapy)
- Rehabilitation of post-stroke aphasia (Master's degree, Speech Language Therapy)
- Aphasia and neuroplasticity (University degree, Neuropsychologic Sciences)
- Prognostic factors in post-stroke aphasia recovery (Research Teaching Unit, Master's degree, Speech Language Therapy)
- Non-Invasive Brain Stimulation in post-stroke aphasia recovery (Master's degree, Speech Language Therapy, University of Geneva)
- Non-linguistic cognitive impairment in post-stroke aphasia: assessment and rehabilitation (Master's degree, Speech Language Therapy)

### **Acquired cognitive disorders (2020-2023)**

- Language and cognitive telerehabilitation (University Degree, Neuropsychologic Sciences)
- Prolonged post-COVID-19 cognitive symptoms in adults: assessment and management (University degree, Neuropsychologic Sciences)

### **Methodology (2021-2024)**

- Bibliographic research, literature review (Bachelor's and Master's degrees, Speech Language Therapy)
- PRISMA and CONSORT grids (Master's degree, Speech Language Therapy and Physical Therapy)
- Theses Preparation Module: finding a research project, formulating a research question, establishing methods, writing a discussion (Master's degree, Speech Language Therapy)
- Preparation for the oral defense of the theses (Master's degree, Speech Language Therapy)

## MENTORING EXPERIENCE

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**Mentored undergraduate** (S. Crouse, S. Zerai, E. Keller) **and graduate** (S. Moore) students on a study of anterior temporal lobe stimulation with transcranial magnetic stimulation to modulate lexicosemantic processes, University of South Carolina.

### **Speech-Language Pathology Theses Supervised**

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| 2026-2027 | Impact of rTMS on brain activity in the recovery of post-stroke aphasia: a systematic review, M. Huberty |
| 2024-2025 | Individualization of rTMS in post-stroke aphasia based on the disconnectome, E. Ferrer.                  |

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|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023-2024 | Effectiveness of Watching Action Verb Videos on Language and Motor Recovery in Post-Stroke Individuals, E. Papin.                                                       |
| 2023-2024 | Telehealth and Speech-Language Therapy in French-Speaking Africa, L. Rabastain.                                                                                         |
| 2022-2023 | Transcranial magnetic stimulation for severe post-stroke patients with aphasia, M. Sanchez.                                                                             |
| 2021-2022 | Transcranial magnetic stimulation in a left-handed patient with aphasia following a right stroke: clinical course and electrophysiological changes, M. de Puy de Goyne. |

**Research internships:** students in speech-language therapy, occupational therapy, psychomotricity, and medicine (France).

## **LANGUAGE AND RESEARCH SKILLS**

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### **Language**

French (native), English (C1), Spanish (B1).

### **Research skills**

Non-Invasive Brain Stimulation:

- Transcranial magnetic stimulation: Magpro, Magventure & BiStim, Magstim.
- Neuronavigation: TMS Navigator & Brainsight.
- Transcranial Direct Current Stimulation: Soterix Medical Inc.

Neuroimaging:

- MRI: Certificate of Completion MRI Level 2 (senior level), USC, record ID: SAH01.
- Analyzing fMRI: AFNI.
- Lesion symptom mapping: NiiStat.
- EEG realization: 128 channels, BioSemi.
- Analysis of Event-Related Potentials: Cartool, Ragu.

Programming and Analysis:

- Vocal analysis: Check-vocal.
- Stimulus presentation software: E-Prime 2.0 & 3.0.
- Programming: R, R-studio.

## **OTHER ACTIVITIES**

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### **Reviewer for international journals**

Journal of Speech, Language, and Hearing Research (2026)

Therapeutics and Clinical Risk Management (2026)

Annals of Physical and Rehabilitation Medicine (2025)  
Brain Structure and Function (2025)  
Aphasiology (2024)  
Neurorehabilitation & Neural Repair (2024)

**Grant review**

Barbara Martin Aphasia Award, Academy of Aphasia and National Aphasia Association (2026)

**Review for national journal**

Glossa (2023)

**Member of the Scientific Committee**

COVIRTUA Healthcare : cognitive rehabilitation platform (2020-2023)

**Theses committee member**

- Speech Language Therapy Theses (2022, 2023, 2024, 2025)
- Neuropsychologic Sciences, University degree (2021, 2022, 2023)
- Medical thesis (2024)

**Local organization team**

International congress 'Science of Aphasia', Bordeaux, September, 2022, with B. Glize & A. Camiade

**Volunteering**

- MOBIL' (student association of international solidarity of Bordeaux), Cofounder and president (2018-2020)
- GOAMA (student association of international solidarity of Nancy)  
Vice president in charge of communication (2015-2016)